Imbalanced Breast Cancer Analysis

RESEARCH BRIEFING & IMPLEMENTATION REVIEW

1. Project Overview & Steps

Objective: Build a robust SVM/DT classifier for a 90:10 Imbalanced Dataset.

Steps Followed:

1. **Data Engineering:**
   * Loaded Breast Cancer Data (Benign vs Malignant).
   * Undersampled Malignant class to force a strict 90:10 imbalance ratio.
2. **Model Configuration:**
   * Implemented SVM (Linear) and Decision Tree.
   * Configured three variants for each: Baseline, Balanced Weights, Custom Weights {0:9, 1:1}.
3. **Validation:**
   * Stratified Split (80/20).
   * Metrics: Recall (Sensitivity), Precision, F1-Score, ROC-AUC.

2. Key Concept: The Imbalance Problem

What it is:

When 'Healthy' patients outnumber 'Sick' patients by 9 to 1.

Why it matters:

Standard algorithms optimize for *Average Accuracy*.

* If we predict "Everyone is Healthy", we get 90% Accuracy.
* But we miss 100% of the Cancer.

Solution:

* **Class Weights: Penalize the model 9x more for missing a cancer case than for a false alarm.**
* **Metric Shift: Move from Accuracy to Recall (Sensitivity).**

3. Execution Output & Results

| Model Variation | Precision | Recall (Key) | F1-Score | Note |

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Presentation

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SVM Baseline	**High**	**Low (~0.0)**	**Low**	**Fails to detect cancer.**
SVM Balanced	**Moderate**	**High (>0.8)**	**High**	**Success.**
DT Baseline	**High**	**Low/Mid**	**Moderate**	**Unstable.**
DT Custom	**Moderate**	**High**	**High**	**Good alternative.**

Observation:

The "Baseline" models fell into the accuracy trap. They ignored the minority class because it was "cheaper" to just predict the majority.

4. Detailed Observations

*   **Recall is the Vital Sign:**
    *   We observed that Weighted SVMs consistently flagged the malignant cases.
    *   Trade-off: This increased False Positives (Healthy people flagged as sick). In oncology, this is acceptable; missing a tumor is not.

*   **Hyperplane Behavior:**
    *   The SVM Baseline pushed the boundary deep into the malignant territory to satisfy the Benign majority.
    *   **The Weighted SVM pushed the boundary back, creating a wider safety margin around the cancer cases.**

5. Conclusion & Recommendation

Final Verdict:

Deploy SVM with Custom/Balanced Weights.

Why?

1. **Safety: It maximizes Recall, ensuring patient safety.**
2. **Robustness: SVM margins generalize better than Decision Trees on small, imbalanced datasets.**
3. **Adjustability: The weight parameter allows doctors to tune the sensitivity preference without retraining.**